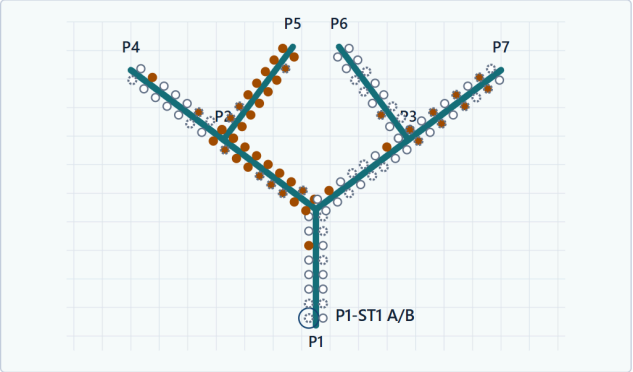
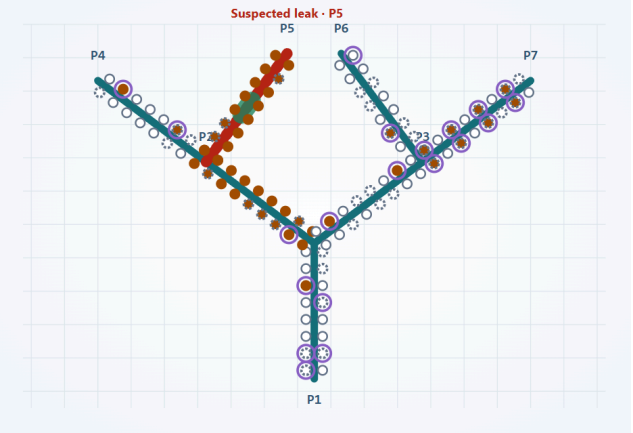
Subsea Integrity Copilot - Customer Brief

The decision. Keep the pressure alerts; improve the investigation that follows. Help an integrity engineer distinguish possible leaks from unreliable instruments, decide where to inspect first, and explain the evidence and uncertainty.

Customer and problem. NorthSea Integrity Operations is a fictional offshore gas operator. Existing detectors flag unusual readings across its pipeline network. Engineers must connect sensor histories, timing, topology, shared gateways and operating records: leaks, sensor faults and operating changes can overlap. An alert is not a diagnosis. This work depends on experienced engineers and is difficult to hand over. Missed leaks risk environmental releases and disruption; false diagnoses can trigger costly vessel or ROV inspections.

Same data, different investigation. Actual dashboard crops below share S7, seed 2026, source +50 to +110 seconds, measurements and detector alerts. The alert-only workflow illustrates the assumed customer process.

Sensor alerts only	Agent-assisted investigation
	
112 instruments; 69 flagged. Orange = pressure alert; dashed outline = missing/invalid. The engineer must compare histories, retrieve records and connect the dots. No combined diagnosis or inspection area.	**An investigation to review.** Red = suspected P5 leak; green = report-stated 700-900 m inspection priority; purple = reported instrument issues. Not a confirmed leak location.

Three questions to test. "Leak or faulty sensors?" "Where should we inspect, and why?" "What supports this, and what remains unknown?" Open measured comparisons, cited records, competing explanations and verification steps. A grounding reviewer checks source support; rejected reports are withheld. Unsupported locations stay withheld. Detector warnings never wait for AI. Detection evaluation uses pre-authored synthetic truth; answer-quality ratings are advisory model opinions.

Buyers and objections. The **Head of Subsea Operations** asks whether effort saved justifies integration, inference and review costs. The **Lead Subsea Integrity & Instrumentation Engineer** asks whether conclusions survive dirty data, conflicting records and reproduction from source. A cheaper fixed workflow may suffice; AI can be wrong; data access and accountability require approval.

Offer and buying gate. A proposed four-week, read-only shadow pilot on one pipeline. Agree manual/fixed-workflow baselines, access controls, spend limits and acceptance thresholds first. Measure engineer review time, missed/false findings, unsupported claims, useful inspection priorities and end-to-end failures/latency. Expand only if evidence quality, effort and total cost justify adoption. No savings or response-time guarantee.

Why this problem. My PhD studied marine data quality, anomaly detection and translating expert requirements into reviewable software.

Limits. Synthetic data, not a validated hydraulic model or field-reliability result. Images show a selected recorded investigation, not a new inference or measured labor saving. Inspection priorities are not dispatch authorization. Production connectors and expert-approved procedures require customer work. Humans retain safety authority.

Demo and run instructions: github.com/ntnguyen-so/subsea-integrity-copilot. Capture evidence.